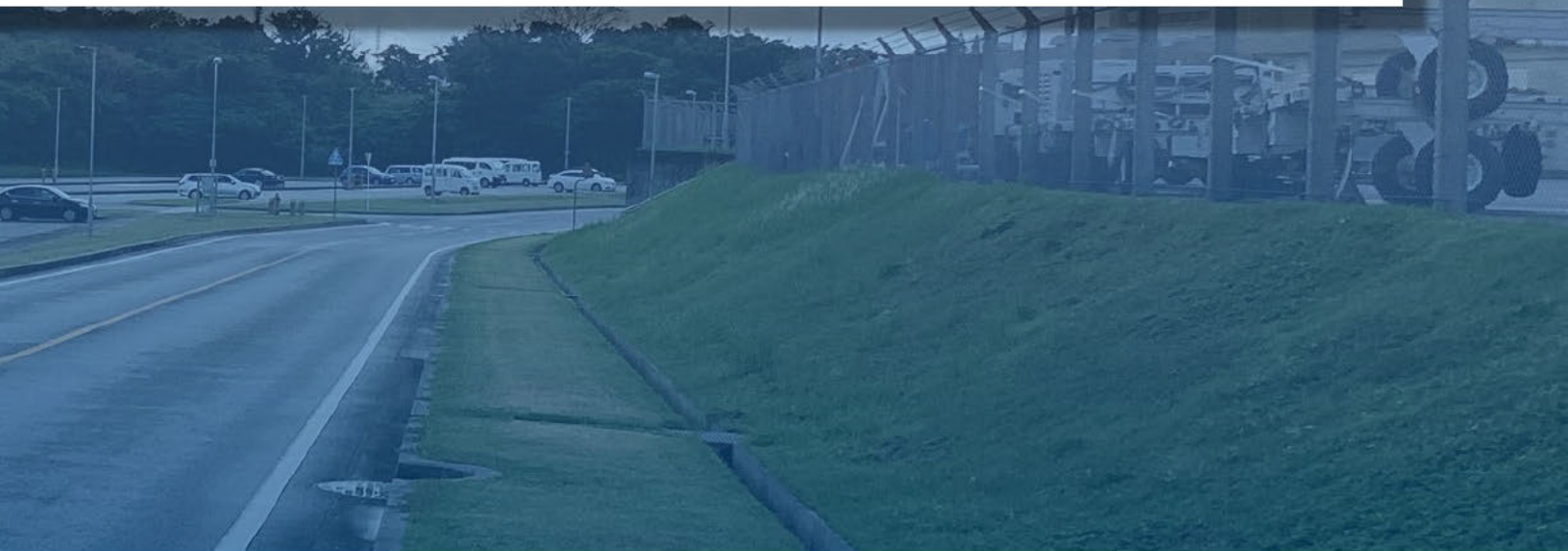


Facility Sustainment, Restoration, and Modernization Charrette Report (CR)

FINAL LXEZ201076CR

CR-Repair Water Lines Around Airfield, Phase II



Kadena Air Base, Japan

MAY 2023

Contract Number: W912BV-19-D-0009

Task Order Number: W912BV21F0077



U.S. AIR FORCE



Michael Baker

INTERNATIONAL

Facility Sustainment, Restoration, and Modernization

FINAL

LXEZ201076CR CR-Repair Water Lines Around Airfield, Phase II

Prepared for:

718 CES/CENPD

Kadena Air Base, Japan

Prepared by:

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Under Contract to:

U.S. Army Corps of Engineers, Tulsa District

Kadena Air Base, Japan

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Michael Baker
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Addendums

Kadena AB Scope Analysis Tracker
AF Form 813 (Air Force generated)



FACILITY SUSTAINMENT, RESTORATION, AND MODERNIZATION (FSRM)

CHARRETTE REPORT (CR)

LXEZ201076CR CR-Repair Water Lines Around Airfield, Phase II

Kadena Air Base, Japan

LXEZ201076CR

MAY 2023

718 CES/CENPD

Point of Contact:

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DSN 315-634-7366

1.0 Introduction

This document provides the Facility Sustainment, Restoration, and Modernization (FSRM) Charrette Report (CR) for the Repair Water Lines Around Airfield, Phase II at Kadena Air Base, Japan.

The primary purpose of the CR is to develop:

- An understanding of Organizational Mission (User) Requirements
- An understanding of Facilities Requirements
- An Analysis of Alternatives
- Project Scope, Siting, and Schedule.
- Draft programming documentation (DD Form 1391 package) in preparation for approval by the proper authorities (Typically AFCEC/CL or SAF/IEE).
- A Class 4 cost estimate for the selected scoping alternative, as well as Class 5 cost estimates for the other scoping alternatives. Cost estimates are to be developed as recommended in AACE PR 56R-08.
- A basis for the future DBB designer to understand the project requirements and establish a framework for project execution.

The CR development was accomplished by Michael Baker International (MBI). Key users included in developing this document included:



The project scope and costs are described below and in the Appendices. The draft DD Form 1391 is provided in **Appendix B**.

The following is a summary of the basic project information:

Project Name: CR – Repair Water Lines Around Airfield, Phase II
Project Number: LXEZ201076CR
Installation: Kadena Air Base
Requiring Command: Pacific Air Force Command (PACAF)
Host Command: Kadena Air Base
Fiscal Year (FY) Design Planned: 2025
Fiscal Year (FY) Construction Planned: 2027
Category Code(s): 842245 (Water Distribution Mains)
Proposed Acquisition Method: Design Bid Build
Status of Design: 15% Design
Design Agent: U.S. Army Engineer District, Japan (USAEDJ)
Construction Agent: USAEDJ

2.0 Project Description

The proposed work in Repair Water Lines Around the Airfield, Phase II primarily focuses on the rerouting and replacement of the existing water mains that are located underneath the airfield runways. Replacing the old deteriorating water mains requires the installation of new fire hydrants along the proposed alignments in accordance with the Unified Facilities Criteria (UFC) and National Fire Protection Association (NFPA) requirements as well as providing water main connections onto existing distribution mains and service lines that feed the current facilities.

The proposed work is led by the replacement and upgrade deteriorating cast iron and asbestos concrete pipe that feeds Reduce Pressure Zone 2 as depicted in the July 2017 Water Distribution System Master Plan prepared by AECOM (hereafter referred to as the Master Plan). As stated in the Master Plan, the water main network that serves Kadena Air Base has a variety of capacity, pressure, and structural integrity issues. The existing water lines are at the end of their life cycle, being over 60 years old, and are prone to high rates of leakage and unplanned ruptures. The original water main network runs through the middle and at both ends of the runways, which puts airfield operations at risk of closure if those segments require repairs.

Additionally, the Master Plan identified several fire protection/potable water demand and pressure deficiencies within the water distribution network. These deficiencies required that the water distribution network be upgraded to accommodate the existing firefighting and potable water demands of the base. The Master Plan proposed multiple water main alignments routed around the airfield along with recommend pipe size and the installation of new pressure reducing valves to balance the water distribution network's pressure and flows within Pressure Zones 1 and 2. The proposed work will provide new PRVs and appropriately sized PVC water mains to ensure balanced pressure and flow throughout the network.

2.1 General Description

The primary purpose of the CR is to validate site conditions to evaluate repair options for Water Line Phase II, develop a proposed project scope and costs, and produce DD 1391 programming documentation that follows the U.S. Air Force's latest guidance and templates.

This project is the first phase in identifying a plan for improving critical and non-critical items, including but not limited to:

- Reduction in number of pressure zones and pressure reducing valves (PRVs)
- Water supply line replacements
- Improved connections to the pump station and reservoir
- Improvements to regulate system pressure

Additional plans beyond this 15% submittal will enhance planning development, timelines, and implementation priorities.

See **Figure 1**, Vicinity Map, and **Figure 2**, Project Site Map for a review of the existing conditions. Conceptual drawings with detailed project descriptions appear in **Section 2.6-Concept Sketches** and in **Appendix C**.

2.2 Current Situation

The existing water distribution system includes water mains ranging from 4-inch to 20-inch in diameter. While a majority of the system has either polyvinyl chloride (PVC) or cast-iron pipes, other water main materials include asbestos cement (AC), ductile iron (DI), galvanized iron (GI), copper, and high-density polyethylene (HDPE). The water main system distributes over 3.6-million gallons of water per day and provides services to over 22,000 people that reside or work on base. The base seeks to optimize its firefighting and potable water distribution network while maintain a maximum water main pressure of 130 psi, throughout its distribution network.

The existing system is over 60-years old, has firefighting demand flow, and pressure issues along the segments that are not structurally sound and may burst and/or leak water at any moment. Additionally, there is currently no SCADA system in place. The Master Plan indicated the following operational concerns:

- Water distribution system has seven existing pressure zones that do not reduce pressure at appropriate locations and exhibit pressures below and above service level criteria.
- The Master Plan revealed the distribution system does not meet operating pressure requirements as outlined in the UFC service level criteria.
- The hydraulic model in the Master Plan indicated fire flow capacities of less than 500 gallon per minute in portions of the flightline.
- Flow measuring test indicate an unusually high amount of nighttime water usage, indicative of a high level of leakage.
- The water distribution system is undersized with approximately 33 percent of the mains of 6-inch or smaller diameter. Smaller diameter water main limits the available flow in the distribution system and contributes to the inadequacy of the system to meet building fire flow requirements.
- Approximately 45 percent of the water mains are cast iron, which was generally believed to have been installed prior to 1960. Older cast iron water mains are likely to be more prone to water main breaks and leakage (see photos on the following page).
- Water Shop personnel indicated that during the past years numerous repairs were necessary due to water main breaks and leakage throughout the air base.



Push on cast iron lead connection beginning to leak at lead joint material on water main (Valve 18).



Leaking gate valve seals on water main (Valve 19).



Push-on lead joint connection on water main is leaking.

2.3 Impact if Not Provided

Failure to maintain the water distribution network can generate several critical issues including high amount of water leakage; low firefighting flows and residual water pressures; high number of emergency leak repairs and unscheduled airfield/roadway closures; and reduce the capacity of the base to provide firefighting demands during emergency situations.

2.4. Alternatives Considered

Alternatives Summary

Alternatives 1 & 2 have been redacted for the purpose of this bid package.

Alternative 3:

This alignment is located within roadway right-of-way, which minimizes work within the airfield and does not impact airfield operations. The proposed water network will connect into the existing water network and buildings as needed. New hydrants will be installed per the UFC.

Alternative 3 (**Figure 5**) is the preferred alternative. This alternative also addresses the critical repairs to the water network but provides the advantage of not impacting airfield operations.

Pros and cons of alternative:

Pro:

- Addresses all critical repairs to water main network.
- Within roadway right of way, minimizes impact to the airfield.
- Cost-effective option construction duration.

Con:

- Does not take advantage of the existing water tank and pump house near Walker Road that can further reduce the need for a pump station.

2.5 REDACTED

2.6 Concept Sketches

The current condition map and proposed work map are included in **Appendix C**. These maps were developed based on recommendations in the Master Plan, this project's 2022 field investigation, and input from stakeholders who participated in the Charrette. Details regarding the recommended work for each alternative are also provided in a subsequent table in this section.



FIGURE 1: VICINITY MAP



FIGURE 2: PROJECT SITE MAP

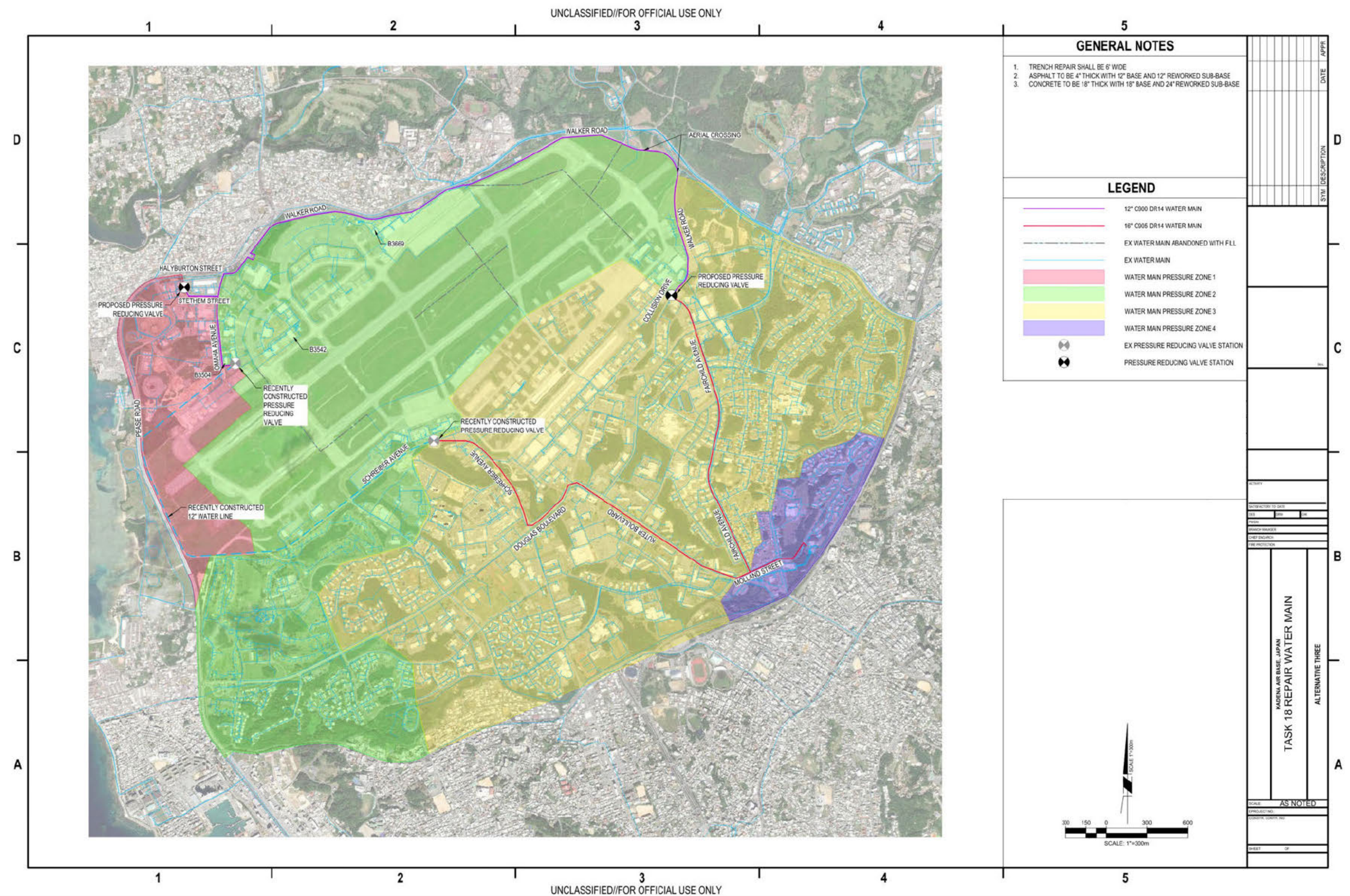
Overall, this project's field investigation and subsequent evaluation determined that the recommendations from the Master Plan are the recommended repair alternative for Water Line Phase II.

Based on the recorded distresses identified in the Master Plan and the 2022 field investigation, the following repairs for the proposed alternatives are considered critical with only **Alternative 1** identifying non-critical repairs. **Figures 3-5** on the following page provide a graphical representation of the work included in each alternative.

Alternative 1	• REDACTED
Alternative 2	• REDACTED
Alternative 3 (preferred) <u>Life, Health, and Safety and Critical Repairs</u> <i>Optimizes Water Line alignment within roadway right-of-way – no impact to airfield operations– Most cost effective</i>	• New pipe network located within roadway right-of-way, minimize work within the airfield with no impact to airfield operations. • Connect onto the existing water network system and building as needed. • New Hydrants per UFC. • Asphalt trench repair. • 12-inch C900 DR 14 water pipe and 16-inch C905 DR 14 water pipe. • Utilizing PRVs to control pressure at the lower areas to regulate system pressure. • SCADA with PRVs to acquire and analyze accurate data.

FIGURE 3: ALTERNATIVE 1 (REDACTED)

FIGURE 4: ALTERNATIVE 2 (REDACTED)



2.7 Charrette Report Development Schedule

LXEZ201076CR Repair Water Lines Around Airfield, Phase II	START DATE	END DATE	CALENDAR DAYS	NOTES
Base Task 18 – Repair Water Lines Around Airfield, Phase II	9/30/2022	4/14/2023	161	Start date based on MOD P00002 Award
RFI Request & Invitations for Mini Workshop	9/30/2022	10/6/2022	7	Issued no more than 7 calendar days following MOD 2 Award
Task 18 Workshop	11/8/22	11/9/2022	2	Held 4 weeks after RFI Request & Invitations complete
Task 18 Revised Draft CR Report and DD1391 (Draft)	11/9/2022	3/9/2023	90	Submitted within 90 calendar days of Workshop
Gov't Review (Task 18 Revised Final)	3/10/2023	3/28/2023	14	Assuming 2-week gov't review period
Task 18 Revised Final CR Report and DD1391 (Final)	3/28/2023	4/14/2023	14	Within 2 weeks of close of gov't comments on Draft and within 104 calendar days of MOD 2 Award
Task 18 Post-Final Revisions/Resubmittal	4/14/2023	5/5/2023	21	Conducted following final review from stakeholders at Kadena.

3.0 Scope Analysis

3.1 Cost and Scope Thresholds

REDACTED

3.2 User and Facility Requirements

General Project Data & Technical Deliverables. The following provides a definition for each, as well as deliverables examples.

General Project Data: These deliverables typically contain project definition that can be developed within a 2% to 10% design effort. Project definition has begun and progressed to at least an intermediate level of completion. Review and approvals for its status has occurred:

- **Project General Scope Description:** The proposed work in Task 18 is primarily focused on the replacement of the water mains throughout the base. Replacing the water main results in additional upgrades surrounding the system. This includes the installation of new fire hydrants along the proposed alignments in accordance with the NFPA as well as providing water main connections onto existing distribution mains and service lines that feed the current facilities.
- **Project Location:** Kadena Air Base, Japan
 - **Total Building Area:** N/A
 - **Functional Space Requirements:** N/A
 - **No. of Building Stories:** N/A
 - **Exterior Closure Description:** N/A
 - **Finishes Descriptions and Requirements:** N/A
 - **Building Code or Standards Requirements:**
 - UFC 3-201-01 Civil Engineering
 - UFC 3-230-01 Water Storage and Distribution
 - UFC 3-600-01 Fire Protection Engineering for Facilities
 - UFC 3-570-01 Cathodic Protection
 - Recommended Standards for Water Works “Ten States Standards”
 - Japan Environmental Governing Standards (JEGS)
- **Fire Protection and Life Safety Requirements:** Fire protection to all buildings and hangers per UFC will be provided.
 - **Security System:** N/A
 - **Seismic Upgrade Requirements:** N/A
 - **Cybersecurity Requirements:** N/A

- **Flood Plain Determination:** The waterlines associated with this project fall within the 100yr floodplain. The Floodplain Management Rule applies.
- **Climate Resilience Requirements (Hurricane prone, lightning prone, flood prone, sea level rise:** N/A
- **Soils and Hydrology Report:** N/A
- **Integrated Project Plan:** N/A
- **Contracting Strategy:** Design-Bid-Build

Escalation Strategy and Basis: REDACTED.

General Project Data: These deliverables typically contain project definition that can be developed within a 10% to 15% design effort. Project definition is advanced, and reviews have been conducted. Development may be near completion with the exception of final approvals:

- Mechanical Systems and Total Capacity: N/A
- Electrical Capacity: N/A
- Communication Systems: N/A
- Anti-Terrorism Force Protection Requirements: N/A
- LEED Certification Level (if applicable): N/A
- Project Master Schedule: N/A
- Work Breakdown Structure: N/A

Technical Deliverables: These deliverables are conceptual and typically developed from 2% to 10% design. Work on the deliverable has begun. Development is typically limited to sketches, rough outlines, or similar levels of early completion:

- Building Codes and Standards: N/A
- Fire Protection and Life Safety Requirements: N/A
- Demolition Plan: N/A
- Utility Plan: N/A
- Site Electrical Plan: N/A
- Site Lighting Plan: N/A
- Site Communications Plan: N/A
- Erosion Control Plan: N/A
- Stormwater Plan: N/A
- Landscaping Plan: N/A
- Exterior Elevations: N/A
- Interior Elevations: N/A
- Interior Sections: N/A
- Partition or Wall Types: N/A
- Finishes: N/A
- Doors: N/A
- Windows: N/A
- Restrooms: N/A
- Furniture Plan: N/A
- Fire Protection Plan: N/A
- Room Layout Plan: N/A

- Foundation Plan: N/A
- Foundation Sections: N/A
- Structural Plan: N/A
- Structural Sections: N/A
- Roof Plan: N/A
- Material, Equipment and Systems Requirements: N/A
- Building Envelope/Moisture Protection/Flashing: N/A
- Mechanical/HVAC Plan: N/A
- Flow Control Diagrams: N/A
- Plumbing Plan: N/A
- One-Line Electrical Diagram: N/A
- Electrical Power Plan: N/A
- Life Safety Plan: N/A
- Equipment Schedule: N/A
- Temporary Facilities: N/A

Technical Deliverables: These deliverables typically are developed from 10% to 15% design. Work on the deliverable is advanced. Interim, cross-functional reviews have usually been conducted. Development may be near completion with the exception of final approvals:

- **Site Plan and Existing Site Plan:** See **Appendix C** for plans.

Not Applicable If any of the following information is readily available and developed into Technical Deliverables, it will augment the charrette's usefulness; yet if not available, it is not required:

- Mechanical/HVAC Details: N/A
- Mechanical/HVAC Schedules: N/A
- Plumbing Details: N/A
- Plumbing Riser Diagram: N/A
- Interior Lighting Plan and Drawings: N/A
- Security Plan and Drawings: N/A
- Emergency Communication Plan and Drawings: N/A
- Lightning Protection Plan and Drawings: N/A
- Motor Control Diagram: N/A
- Lighting Control Diagram: N/A
- Lighting Schedules: N/A
- Electrical/Control Panel Schedule: N/A
- Information Systems/Telecommunication Plan: N/A
- Information Systems/Telecommunication Details: N/A

Not Applicable For tenant related requirements or AF work on non-AF installations, provide existing MOA/MOU or support agreements.

Requirements	Remarks
GENERAL	
As-Built Drawings	GIS data for the existing utilities were provided by the base. GIS data provided accurate location of the network, and confirming the existing layout was possible during a field inspection.

Requirements	Remarks
Site Accessibility Concerns	Work within the airfield will require portions of the airfield to be closed, which will impact the base day-to-day operations. Work within the forest/jungle area will require environmental and cultural resource approval.
Construction Security	N/A
Temporary user space	Contractor will require a two-acre laydown area to store materials and equipment and oversee the construction of the proposed infrastructure improvements.
Facility Addition	N/A
Allied Support Requirements	N/A
SITE, CIVIL, AND GEOTECHNICAL	A subsurface geotechnical investigation is recommended along the proposed water main alignment, along with a detailed topographic survey which includes utility mapping and potholing at all proposed water main/existing utility crossings and tie-in points.
Existing Site Conditions	The existing site has 60-plus-year-old water lines that need repair. Water mains run through the whole base and require different layout considerations with respect to the area's use such as commercial, residential, and airfield spaces.
Facility Siting	Proposed water main should be placed with the roadway right-of-way. Crossing underneath the existing runways or taxiways should be avoided. When required, alignments within the airfield should be located in non-paved or shoulder areas.
Removals/Demolition	Existing water main that will be abandoned should be removed and the pavement or landscaped area restored accordingly. Site demolition will consist of trenching through various paved surfaces and landscaped areas that follow the water main alignment.
Geotechnical Report Required	Yes, geotech report is required. One 10-foot bore every 500-feet along the water main alignment.
Special Soil Conditions	N/A
Roads and Drives	Water main alignment will affect roadways. Department of Transportation (DOT) traffic control will be required for roads. Temporary road closure and lane width reduction will need to be in accordance with the Island's DOT maintenance of traffic criteria.
Vehicle Parking	N/A
Site Work	6-foot Wide trench repair will be required along the proposed layout.
Grading & Storm Water	N/A
UTILITIES	

Requirements	Remarks
Water Distribution	Replace existing water main with pipes ranging from 12-16 inch diameter. Provide a system that maintains MAXIMUM 130 psi at all pressure zones.
Sanitary Sewer	N/A
Natural Gas	N/A
Exterior Electrical	N/A
Exterior Communications	N/A
Cathodic Protection	Cathodic protection required for all metal fittings.
ARCHITECTURAL REQUIREMENTS	
Air Force Corporate Facility Standards (AFCFS)	N/A
Air Force Standard Designs	N/A
Installation Facilities Standards (IFS)	N/A
Space Requirement Analysis	N/A
Finishes (exterior walls, interior walls, ceilings)	N/A
SCIF/VAULT	N/A
Sound Attenuation	N/A
Elevators	N/A
Correction of Architectural Barriers Act (ABA) Deficiencies	N/A
State Historical Preservation Office Review	N/A
STRUCTURAL REQUIREMENTS	
Foundation	N/A
Roof	N/A
Structure	N/A

Requirements	Remarks
- Modification/increase in loads	N/A
- Structural Repairs	N/A
- Seismic Evaluation Required	N/A
MECHANICAL & PLUMBING SYSTEMS	
HVAC system or component modifications	N/A
- Unusual Ventilation	N/A
- Special Heating/Cooling loads	N/A
- Temp/Humidity Control	N/A
Plumbing System or Component Modifications	N/A
FIRE PROTECTION REQUIREMENTS	
Category of Work	New water main will upgrade the existing water distribution network to provide firefighting flows and pressures per UFC 3-600-01. New fire hydrants will be placed in accordance with the UFC.
Fire Safety Deficiencies (FSDs) And Corrective Action Plans (CAPs)	Existing water distribution network lacks adequate firefighting flows and pressures along the north flightline. This will be corrected with the installation of a new water main and pressure reducing valve system.
Changes to facility egress	N/A
Changes to Occupant Density	N/A
Facility houses Mission-Support or Mission-Essential Equipment	N/A
Hangar	N/A
ELECTRICAL SYSTEMS	
Primary Power Services	N/A
Interior Electrical	N/A
Lighting	N/A

Requirements	Remarks
Lightning Protection	N/A
RPIE Generators	N/A
Special User Requirements	N/A
ANTITERRORISM / FORCE PROTECTION	
UFC 4-010-01 Antiterrorism Minimum Standards	N/A
- Unique Occupancy Factors	N/A
- Standoff Distances	N/A
- Window or Glazed Door Replacement	N/A
- HVAC Replacement or Upgrade	N/A
- Mass Notification System (MNS)	N/A
Antiterrorism or Force Protection Requirements Beyond Minimum AT Standards	N/A
SUSTAINABILITY REQUIREMENTS	
UFC 1-200-02, HPSB Requirements	N/A
Compliance With AF Sustainable Design and Development (SDD) Implementing Guidance Memorandum	N/A
Third-Party Certification	N/A
ENVIRONMENTAL REQUIREMENTS AND HAZARDOUS SUBSTANCES	
Environmental Permitting Required	Stormwater Pollution Prevention Permit is required.

Requirements	Remarks
Hazardous Materials Remediation	No known contamination areas were identified during the development of this study. General contractor shall review Kadena AB existing water system drawings to be aware of any asbestos-cement-pipe (ACP) in the project work area. Contractor shall properly manage, handle, remove, and dispose of the ACP according to base requirements. The general contractor is required use a certified asbestos inspector to test for asbestos and/or lead-based paint (LBP) for suspect materials and/or paint which will or may be disturbed in existing infrastructure. Proper planning procedures will be used during the demolition to ensure safety of workers and environment. Proper disposal of any demolition/removal debris is required. If applicable, the contractor will submit a sampling plan for review to SGXB and CEIE. During design and construction, a Safety Data Sheet (SDS) for new paint products will be submitted if scheduled is to be used during construction. The contractor will ensure low volatile paint and primer that does not contain isocyanates and/or lead is used. In addition, the general contractor is also required to obtain prior coordination and approval from SGXB on the use of any equipment containing radioactive material (RAM). Examples include LBP XRF analyzer, soil compaction/density tester. The contractor must coordinate with 18 CES Water Shop if water lines will be interrupted.
Environmental Impact Analysis Process (EIAP)	AF 813 has been completed for this project. Environmental concerns are present. Coordinate and follow requirements for minimizing disturbance and/or mitigation. EIAP is required for this project.
DESIGN AND ACQUISITION OF UNFUNDED REQUIREMENTS	
Furniture Fixtures and Equipment (FF&E) Acquisition	N/A
Existing FF&E Reuse	N/A
Non-RPIE Communications Acquisition	N/A

Requirements	Remarks
Non-RPIE Generators/UPS/Other Backup Power Storage or Capability	N/A
Raised Access Flooring	N/A
TEMPEST / HEMP / RF Shielding	N/A
Intrusion Detection Systems (IDS)	N/A
Closed Circuit TV	N/A
Other User Equipment, Including Special Computer Systems or Support	N/A
AIRFIELDS	
Existing Structure	N/A
Repair Alternatives	Repairs in Alternatives 1 & 2: REDACTED
Evaluation of Excess Pavement Facilities	N/A
Course of Action Analysis	N/A
Lighting and NAVAIDS	N/A
Airfield Waivers	N/A
Phasing and Mission Work- Arounds	N/A
Staging/Laydown	N/A
Base Access	N/A
Design Traffic; Structural and Geometric	N/A

Requirements	Remarks
Tiedowns and Grounding	N/A
Problem Soils	N/A
Frost	N/A
Inlets, Manholes, and Other Utility Structures	N/A
Storm Water	N/A
Aircraft Arresting Systems	N/A

3.3 User Equipment

The items listed below are not critical replacements but may incur longer than expected lead times. They have been added as a precautionary measure.

Description	Quantity	Remarks (including supporting utilities, supply source, and acquisition/installation method, i.e., GF/GI, CF/CI, GF/CI, etc.)
PRV	N/A	CF/CI
Prefabricated Sheds for PRVs	N/A	CF/CI

3.4 Investigations Required

(Not part of this CR development, level of effort is limited to 10%-15 percent design phase)

Description	Required? (Yes / No / N/A)	Scheduled Completion Date
Asbestos / Lead Paint	N/A	
Hazardous Materials	N/A	
Interior Destructive Testing	N/A	
Geotechnical Study	N/A	
Contaminated Soils	N/A	
Topographic Survey	N/A	
Water Pressure / Flow Testing	N/A	
Traffic Study	N/A	
Stormwater Study	N/A	

3.5 Hazardous Substances

Hazardous Substance	Substance Present? (Y / N / Unknown)	Date of Last Survey (N/A if no previous surveys)	Survey Required? (Y/N)	Survey Estimated Completion Date (N/A if no survey required)	Abatement Required? (Y / N / Unknown)	Remarks
Asbestos	Y	N/A	N			
Lead Based Paint	Unknown	N/A	N			
PCB	Unknown	N/A	N			
Radon	Unknown	N/A	N			
Ozone Depleting Substances	Unknown	N/A	N			
Environmental Restoration Site (indicate contaminants of concern in Remarks)	Unknown	N/A	N			
PFOS/PFOA	Unknown	N/A	N			
Other (specify)	Unknown	N/A	N			

3.6 Approvals/Exemptions/Waivers

There are no known waivers to any UFCs or AFIs pertaining to this project. The project as executed in Alternative 3 will meet all code requirements.

3.7 Environmental Requirements

An Environmental and Cultural Resource Site Assessment is required.

4.0 Acronyms

Acronym	Definition	Acronym	Definition
ABA	Architectural Barriers Act	MOU	Memorandum of Understanding
AFCFS	Air Force Corporate Facility Standards	NAVAID	Navigational Aid
AFMAN	Air Force Manual	NAVFAC	Naval Facilities Engineering Systems Command
AFI	Air Force Instruction	NFPA	National Fire Protection Association
AT/FP	Antiterrorism/Force Protection	PACAF	Pacific Air Force Command
ACP	Asbestos Cement Pipe	PACES	Parametric Cost Engineering System
BCI	Building Condition Index	PCB	Polychlorinated Biphenyls
CAPs	Corrective Action Plans	PCR	Planning Charrette Report
CF/CI	Contractor Furnished/Contractor Installed	PFOA	Perfluorooctanoic Acid
COA	Course of Action	PFOS	Perfluorooctane Sulfonate
CR	Charrette Report	PRV	Pressure Reducing Valve
DBB	Design-Bid-Build	PSI	Pounds per Square Inch
DI	Ductile Iron	PVC	Polyvinyl Chloride
DOD	U.S. Department of Defense	SCADA	Supervisory Control and Data Acquisition
DOT	Department of Transportation	SCIF	Sensitive Compartmented Information Facility
DR	Drain	SDD	Sustainable Design and Development
EA	Economic Analysis	UFC	Unified Facilities Criteria
ELAP	Environmental Impact Analysis Process	USAEDJ	U.S. Army Engineer District, Japan
FF&E	Furniture Fixtures and Equipment (FF&E) Acquisition	VFD	Variable Frequency Drive
FSDs	Fire Safety Deficiencies		
FSRM	Facility Sustainment, Restoration and Modernization		
FY	Fiscal Year		
GI	Galvanized Iron		
GIS	Geographic Information System		
GPM	Gallons per Minute		
HDPE	High-density Polyethylene		
HOOH	Home Office Overhead		
HPSB	High-Performance Sustainable Building		
IFS	Installation Facilities Standards		
JEGS	Japan Environmental Governing Standards		
JOOH	Job Office Overhead		
LEED	Leadership in Energy and Environmental Design		
LF	Linear Feet		
MBI	Michael Baker International		
MNS	Mass Notification System		
MOA	Memorandum of Agreement		

APPENDICIES

A. List of Participants

B. REDACTED

C. Maps/Plats/Utility/Demolition/Site Sketches/Plans

D. Photos

E. REDACTED

F. Updated Project Scoring Worksheet – NOT APPLICABLE

G. REDACTED

H. Project Schedule

I. AFMAN 32-1084 Space Analysis with Facility Space Optimization Worksheet – NOT APPLICABLE

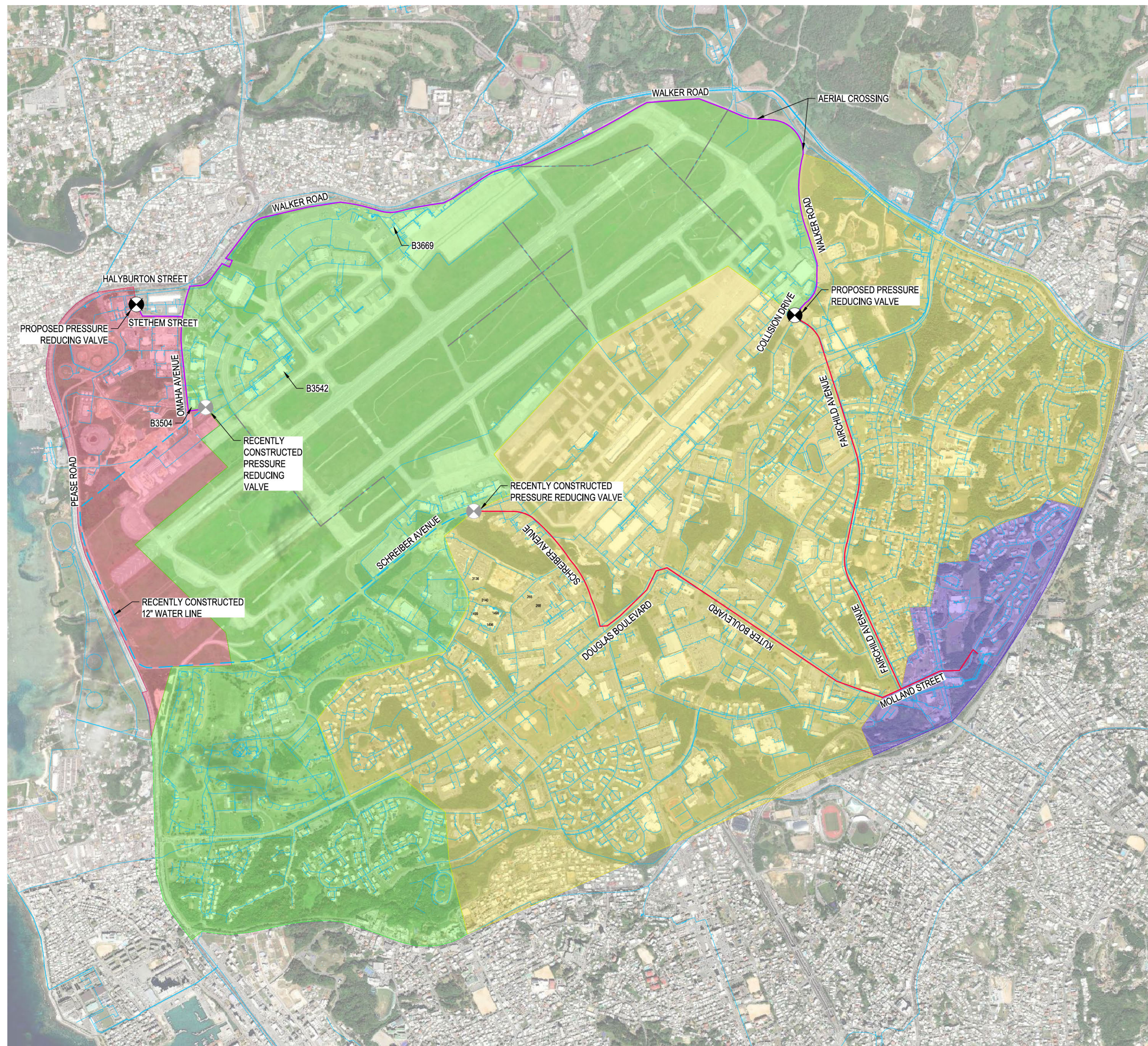
J. Simplified Analysis of Alternatives (SAoA) (not required if the CPP Planning Action (Vector Check) Process is used) – NOT APPLICABLE

K. Draft Economic Analysis (EA)

L. MOA/MOU or Support Agreement – NOT APPLICABLE

A. List of Participants

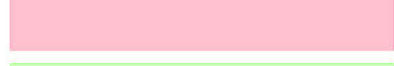
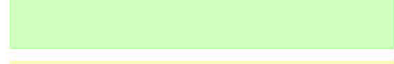
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GENERAL NOTES

1. TRENCH REPAIR SHALL BE 6' WIDE
2. ASPHALT TO BE 4" THICK WITH 12" BASE AND 12" REWORKED SUB-BASE
3. CONCRETE TO BE 18" THICK WITH 18" BASE AND 24" REWORKED SUB-BASE

LEGEND

- | | |
|---|------------------------------------|
|  | 12" C900 DR14 WATER MAIN |
|  | 16" C905 DR14 WATER MAIN |
|  | EX WATER MAIN ABANDONED WITH FILL |
|  | EX WATER MAIN |
|  | WATER MAIN PRESSURE ZONE 1 |
|  | WATER MAIN PRESSURE ZONE 2 |
|  | WATER MAIN PRESSURE ZONE 3 |
|  | WATER MAIN PRESSURE ZONE 4 |
|  | EX PRESSURE REDUCING VALVE STATION |
|  | PRESSURE REDUCING VALVE STATION |

QUANTITIES

ITEM	QUANTITY	UNITS
WATER MAIN ALIGNMENT	38,400	LF
WATER MAIN TRENCH REPAIR - ASPHALT	230,400	SF
WATER CONNECTIONS	150	VALVES
FIRE HYDRANTS	119	HYDRANTS
GATE VALVES	64	VALVES

[illegible][illegible]

SATISFACTORY TO DATE		
DES	DRW	CHK
PMDM		
BRANCH MANAGER		
CHIEF ENGINEER		
FIRE PROTECTION		

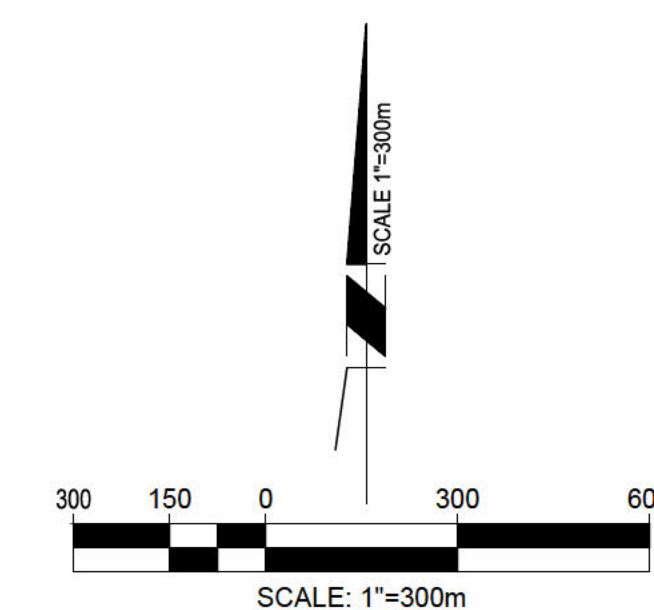
KADENA AIR BASE, JAPAN

TASK 18 REPAIR WATER MAIN

ALTERNATIVE THREE

SCALE:	AS NOTED
EPROJECT NO.:	
CONSTR. CONTR. NO.	

SHEET OF



D. Photos



PHOTO 1: WATER MAIN - Push on cast iron lead connection beginning to leak at lead joint material on water main (Valve 18).



PHOTO 2: WATER MAIN - Leaking gate valve seals on water main (Valve 19).



PHOTO 3: WATER MAIN - Leaky push-on lead joint on water main.

Addendums

KADENA AB Scope Analysis Tracker - Task 18 Repair Water Lines Around Airfield, Phase II

Instructions: Use this page to answer scope questions about each alternative. Some of the questions are driven by the infrastructure systems selected on the scope alternatives page. For the applicable alternative(s), answer these infrastructure-driven questions. If the infrastructure-driven questions are not applicable to all alternatives, "N/A" or blank can be used.

Alternative:	ALTERNATIVE 1: REDACTED	ALTERNATIVE 2: REDACTED	ALTERNATIVE 3: Life, Health, and Safety and Critical repairs - Optimizes Water Line alignment within roadway right-of-way - no impact to airfield operations- Most cost effective
Description:			This alignment is located within roadway right-of-way which minimizes work within the airfield and does not impact airfield operations. The proposed water network will connect into the existing water network and buildings as needed. New items include: hydrants installed per the UFC; asphalt trench repair; 12" C900 DR 14 water pipe and 16" C905 DR 14 water pipe; two new PRVs and appropriately sized PVC water mains to ensure balanced pressure and flow throughout the network; and SCADA with PRVs to acquire and analyze accurate data.
1: GENERAL			
I: What scheduled repair activities are required to maintain or repair the affected real property assets for their expected service life? Check Systems in SMS that need maintenance or repair and service calls. Are they included in this alternative?			Approximately 13 preventive maintenance tasks are currently in place to maintain and restore its functionality. Other work tasks assigned to CEN for project execution such as improvements to water distribution/pressure, leak repairs, corroded pipes/hydrant replacements are all addressed in this alternative. This will also increase the overall utility condition index (UCI 50) of Kadena's water distribution.
II: What critical systems in need of repair/replacement (electrical, HVAC, fire protection, etc.)? Are they included in this alternative?			The proposed work in Task 18 is led by the replacement and upgrade of the existing potable water distribution network. The water main network that serves Kadena Air Base has a variety of capacity, pressure, and structural integrity issues. The existing water lines are at the end of their life cycle, being over 60 years old. The original watermain network runs through the middle of both runways which puts airfield operations at risk of closure if those segments require repairs. This alternative fully addresses these issues.
III: Are five or more key systems in need of repair/replacement? If so, explain whether or not this alternative is a total renovation, repair or replacement.			This project addresses one system, the water distribution network. This alternative is a replacement of the water mains throughout the base.
IV: What siting concerns (traffic, orientation, airfield criteria, etc.) could impact facility design, cost or scope? Explain how this alternative addresses them.			The existing watermain alignment runs under the airfield. Proposed alternatives will either affect airfield operations or impact surface roadways. Demolition and removal of old pipeline may affect airfield operations.
V: What other similar or related functions could be consolidated in this facility or on this site? Explain how this alternative addresses them.			None. This is a replacement of the watermain throughout the installation.
VI: What are the other code compliance, regulatory issues, or current design standards that should be considered? Are they resolved in this alternative?			The watermain does not meet UFC or NFPA requirements. This alternative as executed will meet all code requirements, including UFC 3-201-01 Civil Engineering, UFC 3-230-01 Water Storage, Distribution, and Transmission, and UFC 3-600-01 Fire Protection Engineering for Facilities.
VII: What are the AT/FP requirements for this facility or site? How does this alternative address them? Consider setbacks, glazing requirements, changes in occupancy levels, etc.			Not applicable.

VIII: Would this alternative likely be phased to minimize disruption to occupants or completed at once to minimize impact to mission? Please explain.

IX: What swing space, temporary leases or facilities, or other support for facility occupants or users would be required during the project?

X: If Facilities Board approval is required, include the approval status/date and upload supporting documents (minutes, slides, etc.).

2: Electrical System (Additional Infrastructure System)

I: What are the known deficiencies in primary power services at this site or facility? Not applicable.

II: What are the known deficiencies in existing interior electrical at this facility? Not applicable.

III: What is the current condition of the facility's electrical system? How does this alternative impact it? Not applicable.

IV: What is the condition of the facility's lighting and electrical fixtures? How does this alternative impact it? Not applicable.

V: Is the lightning protection system code compliant based on NFPA 780? How does this alternative impact it? Not applicable.

VI: Is the generator RPIE or equipment? How does that impact this alternative? Not applicable.

VII: What are the special user requirements such as voltage/current and frequency source? Not applicable.

3: Facility Site Utilities (Additional Infrastructure System)

I: What are the known water distribution deficiencies at this site that could impact costs?

II: What are the known sanitary sewer deficiencies at this site that could impact costs?

III: What are the known natural gas deficiencies at this site that could impact costs?

IV: What are the known exterior electrical deficiencies at this site that could impact costs?

V: What are the known exterior communication deficiencies at this site that could impact costs?

VI: What are the known cathodic protection deficiencies at this site that could impact costs?

4: Fire Protection (Additional Infrastructure System)

I: What Risk Assessment Codes (RACs) and/or Fire Safety Deficiencies (FSDs) are assigned to this facility? How are they being addressed in this alternative?

II: What are the anticipated facility egress requirements (capacity, number, arrangement) and how are they being addressed?

III: What are the known natural gas deficiencies at this site that could impact costs?

5: Mechanical & Plumbing Systems (Additional Infrastructure System)

I: What additions or upgrades to the HVAC system or components are needed? Address any issues with ventilation or any special heating/cooling loads.

II: What special requirements are included for industrial operations?

This alternative would likely be phased to minimize impact to mission. Work sequencing plan will be done during the design stage. This should be coordinated and approved by 18OSS/OSAA and 18SFS/55.

Contractor will require a two-acre laydown area to store materials and equipment and oversee the construction of the proposed infrastructure improvements.

Associated equipment such as PRV, fire hydrants and the like will require siting, and therefore needs Facilities Board approval prior to construction.

Not applicable.

Not applicable.

Not applicable.

Not applicable.

Not applicable.

Not applicable.

Not applicable.

The existing water distribution system is over 60-years old, has firefighting demand flow and pressure issues along with segments that are not structurally sound and may burst and/or leak water at any moment. This project replaces existing water main with pipes ranging from 12" -16" in diameter and provides a system that maintains maximum 130 psi at all pressure zones.

None.

None.

None.

None.

Cathodic protection required for all metal fittings.

No RAC/FSDs are assigned to Kadena's water distribution system.

Not applicable.

Not applicable.

Not applicable.

Not applicable.

6: Site, Civil, and Geotechnical (Additional Infrastructure System)

I: What are the existing site conditions that could impact the cost of this alternative?

The existing site has 60+ year old water lines that need repair. Water mains run through the whole base and require different layout considerations with respect to the area's use such as commercial, residential, and airfield spaces. A subsurface geotechnical investigation is recommended along the proposed water main alignment, along with a detailed topographic survey which includes utility mapping and potholing at all proposed watermain/existing utility crossings and tie-in points. For the geotechnical report, one 10-foot bore every 500-feet along the watermain alignment will be performed. Additionally, 6' wide trench repair will be required along the proposed layout.

II: What are the environmental concerns that could impact cost or timeline? Consider LBP. Asbestos, Endangered Species, Historical Significance, Land Use Controls, etc.

Environmental concerns are present (refer to AF 813). During design, general contractor shall review Kadena AB existing water system drawings to be aware of any asbestos-cement-pipe (ACP) in the project work area. Contractor shall properly manage, handle, remove, and dispose of the ACP according to base requirements.

The general contractor is required use a certified asbestos inspector to test for asbestos and/or lead-based paint (LBP) for suspect materials and/or paint which will or may be disturbed in existing infrastructure. Proper planning procedures will be used during the demolition to ensure safety of workers and environment.

During design and construction, a Safety Data Sheet (SDS) for new paint products will be submitted if scheduled is to be used during construction. The contractor will ensure low volatile paint and primer that does not contain isocyanates and/or lead is used.

In addition, the general contractor is required to obtain prior coordination and approval from SGXB on the use of any equipment containing radioactive material (RAM). Examples include LBP XRF analyzer, soil compaction/density tester.

The contractor must coordinate with 18 CES Water Shop if water lines will be interrupted.

Environmental and Cultural Resource Site Assessment is required.

III: What are the requirements for roads/drives, vehicle parking, and sitework that could impact this alternative?

Water main alignment will affect roadways. DOT traffic control will be required for roads. Temporary road closure and lane width reduction will need to be in accordance with the Island's DOT maintenance of traffic criteria.

IV: What are the existing or proposed grading and stormwater issues on the site? How could they impact this alternative?

Stormwater Pollution Prevention Permit is required.

7: Structures & Finishes (Additional Infrastructure System)

I: What are the known deficiencies with finishes (exterior walls, interior walls, ceilings) at this site or facility that could impact costs?

Not applicable.

II: What are the known SCIF/Vault design requirements at this site or facility that could impact costs?

Not applicable.

III: What are the known sound attenuation requirements at this site or facility that could impact costs?

Not applicable.

IV: What are the known elevator repair or replacement requirements at this site or facility that could impact costs?

Not applicable.

V: What are the known Architectural Barrier Act (ABA) deficiencies at this site or facility that could impact costs?

Not applicable.

VI: What are the known roof requirements that could impact costs? If more than 50% of the roofing is being replaced and wind speeds are greater than 115 mph, reach out to AFCEC for confirmation of requirements.

Not applicable.

8: Proposed Work (Scope Analysis)

Response

Alternative 3's northern alignment maintains the proposed 12" C900 DR 14 PVC loop within the roadway right of way which removes all of the construction activities off of the airfield, greatly reducing impact to airfield operations. This alignment begins at the Tank Farm with a 16" C900 DR 14 PVC water main and travels northeasterly along Fairchild Ave to the newly constructed PRV assembly north of Building 938; the pipeline is then downsized to a 12" C900 DR 14 PVC watermain which loops around the runways along Walker Rd to Omaha Ave and ties onto the newly installed southern watermain alignment south of Building 3504; installs a new 12" PVC branch along Stethem St, and installs a pressure reducing valve (PRV) onto the existing water distribution network east of Building 3574. Alternative 3's western alignment begins at the Tank Farm with a 16" C900 DR 14 PVC watermain that travels northeasterly along Kuter Blvd, Douglas Blvd, and Schreiber Ave, and terminates at the newly installed PRV assembly south of Building 3400.

9: Environmental Impact Analysis (Scope Analysis)

Response

AF 813 for this requirement was completed on 29 Oct 2021.

10: Economic Analysis Status (Scope Analysis)

Response

EA is provided.

11: Estimated Year of Execution (Scope Analysis)

Year

2027

12: Design Standards (Scope Analysis)

Response

This project will use all applicable DoD, Air Force, and Kadena design standards.

13: Pros (Scope Analysis)

Response

Fully addresses critical repairs to the watermain network. The least expensive alternative. Minimizes work within the airfield and does not impact airfield operations. Short construction duration.

14: Cons (Scope Analysis)

Response

Does not take advantage of existing water tank and pump house near Walker Rd to reduce the need for a pump station.

15: REDACTED

16: Other Considerations (Scope Analysis)

Response

No other considerations.

17: Preferred Scope Alternative

Preferred Option? Y/N

Yes.

REQUEST FOR ENVIRONMENTAL IMPACT ANALYSIS		Report Control Symbol (RCS): 21-077
INSTRUCTIONS: Section I to be completed by Proponent; Sections II and III to be completed by Environmental Planning Function. Continue on separate sheets as necessary. Reference appropriate item number(s).		
SECTION I - PROPONENT INFORMATION		
1. TO (Environmental Planning Function) 718 CES/CEIE	2. FROM (Proponent organization and functional address symbol) Air Force - 718 CES/CENPD	2a. TELEPHONE NO. (315) 634-1028
3. TITLE OF PROPOSED ACTION REPAIR WATER LINES AROUND AIRFIELD, II (LXEZ201076)		
4. PURPOSE AND NEED FOR ACTION (Identify decision to be made and need date) 1. What do you intend to accomplish and why is the action necessary? PROVIDE RELIABLE WATER SUPPLY NORTHSIDE of KADENA AB 2. What is currently being done to meet the need? Maintain current water distribution system, using urgent and emergency job orders to respond to problems. 3. Provide any additional details related to the Purpose and Need for Action. The water distribution system that brings water to the north side of Kadena AB is extremely old and undersized for the current demand. This project will also relocate the water lines from under the runways so that is the event of a water break it does not jeopardize the airfield. Need Date: 03/01/2022		
5. DESCRIPTION OF PROPOSED ACTION AND ALTERNATIVES (DOPAA) (Provide sufficient details for evaluation of the total action.) 1. What other alternatives have been considered (to include the no action alternative)? STATUS QUO REPAIR WATER LINES AROUND AIRFIELD, II (LXEZ201076) Construct new water system GOJ Replacement of water lines 2. What alternatives were eliminated from consideration and why? STATUS QUO - The base will continue to operate using the existing substandard and deteriorating water distribution system, LS 7401 – KAB Water Distribution Main. Construct new water system - High cost and will replace sections of the water system that are not in bad shape. Water lines will still have to run around both ends of the airfield to provide a continuous loop and prevent dead ends. Wait time for execution is significantly longer than that of renovation. GOJ Replacement of water lines - The time line for the Government of Japan to do this type of repair is several years. There has been a GOJ funded sewer line replacement going on here for nearly 20 years, this work would probably be handled very similarly. 3. Please provide a description of the construction action and timing when it will occur. Repair the water lines running around the east end of the airfield with new water lines that are properly sized for the current load. This project will reroute water lines away from runway and maintain water pressure at maximum 130 PSI throughout the entire water supply system. The work of this alternative will include removal of the existing waterlines. FY26 4. Describe the project location. Attach map(s)/diagram(s) – make sure to include an overview map of where your requested project area is on the installation. Around the Airfield. 5. Describe additional project requirements: 1) Construction and site preparation requirements (include approx. area of ground to be disturbed); 2) Does the project require a laydown yard or storage area? If so, describe the location and groundwork required. Currently there are two main lines support the north side water supply. These two mainlines runs through flight lines, routine water line maintenance requires partial runway shut down. Due to friction and bend angle turns, water pressure drop is unavoidable, especially around building 7401 area, water pressure is around 40 PSI, in case of fire outbreak and there is not enough water pressure, fire damage can significantly increase. 6. Describe additional project requirements: 3) Will soil boring/sampling/potholing occur during a design phase? If so, a separate dig permit will be required; 4) Detailed operational activities; 5) Equipment/material lists. ACM/LBP sampling and testing, and possibly soil boring. 7. Provide any additional details related to the Description of the Proposed Action and Alternatives. Map Attachments: Waterline map.png Location Description / Justification: LS 7401 – KAB Water Distribution Main		
6. PROPONENT APPROVAL (Name and Grade) Okuhara Alex DOD - alex.okuhara	6a. SIGNATURE //Okuhara Alex DOD - alex.okuhara i:0e.t[fedvis]alex.okuhara//	6b. DATE 08/17/2021
SECTION II - PRELIMINARY ENVIRONMENTAL SURVEY (Check appropriate box and describe potential environmental effects including cumulative effects.) (+ = positive effect; 0 = no effect; - = adverse effect; U = unknown effect)		+ 0 - U
7. AIR INSTALLATION COMPATIBLE LAND USE/ZONE USE (Noise, accident potential, encroachment, etc.)		X
8. AIR QUALITY (Emissions, Attainment status, state implementation plan, etc.)		X
9. WATER RESOURCES (Drinking water, wastewater, quality, quantity, source, water features, etc.)		X
10. SAFETY AND OCCUPATIONAL HEALTH (Asbestos/lead-based paint/radiation/chemical exposure, explosives safety quantity distance, bird/wildlife aircraft hazard, etc.)		X
11. HAZARDOUS MATERIALS/WASTE (Use/storage/generation, solid waste, toxic materials, etc.)		X
12. BIOLOGICAL RESOURCES (Wetlands/floodplains, threatened or endangered species, etc.)		X
13. CULTURAL RESOURCES (Burial sites, archaeological, historical, etc.)		✓

		A	
14. GEOLOGY AND SOILS (Topography, minerals, geothermal, Installation Restoration Program, seismicity, etc.)		X	
15. SOCIOECONOMIC (Employment/population projections, school and local fiscal impacts, etc.)		X	
16. OTHER (Potential impacts not addressed above, such as Host Nation considerations/concerns for non-US locations.)		X	
SECTION III - ENVIRONMENTAL ANALYSIS DETERMINATION			
17. ☐ a) PROPOSED ACTION IS NOT A MAJOR FEDERAL ACTION, THEREFORE, NO FURTHER ENVIRONMENTAL IMPACT ANALYSIS IS REQUIRED. (SEE REMARKS IN BLOCK 18); OR ☐ b) PROPOSED ACTION QUALIFIES FOR AN EXEMPTION (SEE REMARKS IN BLOCK 18); OR ☐ c) PROPOSED ACTION WILL NOT CAUSE SIGNIFICANT HARM TO THE ENVIRONMENT OF THE GLOBAL COMMONS, SO NO FURTHER ENVIRONMENTAL IMPACT ANALYSIS IS REQUIRED. SUPPORTING ENVIRONMENTAL ASSESSMENT OR COMPLETED SIGNIFICANT HARM ASSESSMENT (SHA) FOR EIAP OUTSIDE THE U.S. (EIAP-O) CHECKLIST IS ATTACHED. (SEE REMARKS IN BLOCK 18); OR ☒ d) PROPOSED ACTION WILL NOT CAUSE SIGNIFICANT HARM TO THE ENVIRONMENT OF A FOREIGN NATION OR TO A PROTECTED GLOBAL RESOURCE. SUPPORTING SHA FOR EIAP-O CHECKLIST (IF COMPLETED) IS ATTACHED. (SEE REMARKS IN BLOCK 18); ☐ e) PROPOSED ACTION WILL CAUSE SIGNIFICANT HARM TO THE ENVIRONMENT OF A FOREIGN COUNTRY OR A PROTECTED GLOBAL RESOURCE REQUIRING FURTHER ENVIRONMENTAL ANALYSIS IN THE FORM OF AN NVIRONMENTAL REVIEW (ER) OR ENVIRONMENTAL STUDY (ES). (SEE REMARKS IN BLOCK 18); OR ☐ f) PROPOSED ACTION WILL NOT CAUSE SIGNIFICANT HARM TO THE ENVIRONMENT OF A FOREIGN COUNTRY OR A PROTECTED GLOBAL RESOURCE, BUT FURTHER ENVIRONMENTAL ANALYSIS IN THE FORM OF AN ENVIRONMENTAL REVIEW (ER) OR ENVIRONMENTAL STUDY (ES) IS APPROPRIATE ACCORDING TO LEADERSHIP OR THE EPF. (SEE REMARKS IN BLOCK 18); OR ☐ g) PROPOSED ACTION WILL CAUSE SIGNIFICANT HARM TO THE ENVIRONMENT OF THE GLOBAL COMMONS REQUIRING FURTHER ENVIRONMENTAL ANALYSIS IN THE FORM OF AN OUTSIDE THE U.S. ENVIRONMENTAL IMPACT STATEMENT (OEIS). (SEE REMARKS IN BLOCK 18); OR ☐ h) ENVIRONMENTAL IMPACT ANALYSIS IS REQUIRED PER THE TERMS OF AN INTERNATIONAL AGREEMENT. (SEE REMARKS IN BLOCK 18).			
18. REMARKS The proposed action is a continuation or resumption of preexisting actions, where there is no substantial change in existing conditions or existing land uses and where the actions were originally evaluated in accordance with applicable law and regulations, and surrounding circumstances have not changed.			
19. ENVIRONMENTAL PLANNING FUNCTION CERTIFICATION (Name and Grade) Chen, Chinling GS-12		19a. SIGNATURE //Chen Chinling DOD - chinling.chen i:0e.t[fedvis]chinling.chen//	19b. DATE 10/29/2021

CONTINUATION SHEET
Review Comments: Water Resources (08/19/2021 - Westerlund Dacia DOD - dacia.h.westerlund) Ensure sufficient erosion and storm water runoff control measures are in place that meet the installations Storm Water Pollution Prevention Plan to reduce erosion. Cultural Resources (09/09/2021 - Cobos Hugo DOD - hugo.a.cobos) Cultural Asset Survey (CAS) will be required to ensure project will not adversely affect unknown and known cultural assets. Project will go through areas that have not been surveyed that will need to be cleared prior to work beginning. Some lines also go through known sites. These lines will either need to be rerouted around the features or consultation and data recovery with the GOJ will be required prior to work beginning. Biological Resources (09/09/2021 - Cobos Hugo DOD - hugo.a.cobos) Banyan trees located along the project lines. Ensure these are not damaged during implementation by rerouting lines or relocation of the trees. Contact 718 CES/CEIE if further action is required or for alternatives to relocation. Legal (10/22/2021 - Poelma Richard DOD - richard.d.poelma1) The AF813 is legally sufficient as it demonstrates the proposed action is a continuation or resumption of preexisting actions, where there is no substantial change in existing conditions or existing land uses and where the actions were originally evaluated in accordance with applicable law and regulations, and surrounding circumstances have not changed. Plans to mitigate risk of disturbance of cultural resources has been noted. Attachments: